BIOCHEMISTRY CUMULATIVE EXAM

January 20,1996

The enzyme phosphofructokinase (PFK) has been called the "pacemaker" of glycolysis. It catalyzes the reaction

fructose-6-phosphate + ATP → fructose-1,6-bisphosphate + ADP.

The activity of PFK in both liver and muscle is stimulated by fructose-2,6-bisphosphate (F-2,6-BP). The concentration of F-2,6-BP is under hormonal control. However, the hormonal regulation of F-2,6-BP in liver and muscle is quite different.

1. (30) Briefly outline the response of glycogen metabolism and glycolysis to an emergency which results in the release of epinephrin (adrenalin) into the blood stream. Discuss the different roles of muscle and liver in responding to this emergency.

2. (30) Describe the different responses of PFK in liver and muscle to the above emergency which triggers the release of epinephrin. Discuss why these different responses are necessary for the organism.

3. (40) Describe the overall mechanism by which epinephrin release regulates the concentration of F-2,6-BP, and hence the activity of PFK. Discuss how this mechanism differs between liver and muscle.